

Kalman Delta Networks 3: Mixture of Associative Memories

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Abstract

Kalman Associative Memory models a recurrent key–value map as one latent matrix with a columnwise Gaussian filtering law. We seek to scale its state capacity by maintaining many Gaussian associative memories while reading and writing only a small token-selected subset. The derivation proceeds in three stages. First, replacing the single Gaussian by a persistent finite Gaussian sum gives an exact Bayesian anchor: linear–Gaussian prediction and conditioning preserve each hypothesis, component evidence updates its probability, and the one-component boundary is precisely Kalman Associative Memory. Second, allowing a fresh categorical assignment at every token defines a conditional mixture of dynamic Bayesian linear regressors. Exact filtering then branches over routing histories. An online structured mean-field projection compresses this posterior to a product of Gaussian memory banks and yields responsibility-weighted, fractional Kalman updates. Restricting its categorical factor to a token-local recognition network makes all routing coefficients known before the recurrent scan; it is an amortized variational filter when trained against the corresponding objective, and a variationally derived gate otherwise.

Third, a deterministic top- k participation controller defines a sparse, local-event-time member of the same model family. Inactive memories then have identity dynamics and zero process noise, so routed tokens can be packed into expert-specific chronological subsequences and processed by segmented KDA/KDN scans. The number of stored memories grows with the component count C , while expensive recurrent work grows with k rather than C ; a global-time-aging alternative is also characterized. This note therefore separates the exact Gaussian-sum motivation, the approximate variational model, and the sparse systems realization. Current multi-head KDN is recovered at $C = 1$ after imposing the same covariance approximation.

1 Scope and notation

We start directly from the Kalman Associative Memory formulation. The latent memory $\tilde{\mathbf{S}}_t \in \mathbb{R}^{d_k \times d_v}$ is a random linear map from keys to values. A tilde denotes a random state, a bare symbol denotes a filtered component mean, and a hat denotes an exclusive one-step prediction before token t is assimilated. For matrices $\mathbf{X} = [\mathbf{x}_1, \dots, \mathbf{x}_{d_v}]$ and $\mathbf{M} = [\mathbf{m}_1, \dots, \mathbf{m}_{d_v}]$, retain the columnwise Gaussian notation

$$\mathbf{X} \sim \mathcal{N}_{\text{col}}(\mathbf{M}, \mathbf{P}) \iff \mathbf{x}_j \sim \mathcal{N}(\mathbf{m}_j, \mathbf{P}) \text{ independently for } j = 1, \dots, d_v. \quad (1)$$

Equivalently, $\text{vec}(\mathbf{X}) \sim \mathcal{N}(\text{vec}(\mathbf{M}), \mathbf{I}_{d_v} \otimes \mathbf{P})$.

Let C denote the number of mixture components and let $c \in \{1, \dots, C\}$ index them. We use C rather than the commonly used E to avoid confusing the component count with expectation. The base construction has one categorical variable Z shared by every value column of a head and persistent through the sequence. This shared label is essential: conditional on Z , the columns are independent,

but marginalizing Z can create dependence between them. Independent component labels for the d_v columns would instead define a different model with as many as C^{d_v} joint combinations.

The persistent label provides a simple setting in which an exact finite Gaussian mixture remains closed. It represents alternative posterior hypotheses about one latent memory. It does *not* yet represent a mode that switches between tokens or a bank in which only one memory is updated. Those models are separated in Section 8.

2 Kalman Associative Memory

The latent associative map evolves according to

$$\tilde{\mathbf{S}}_t = \mathbf{D}_t \tilde{\mathbf{S}}_{t-1} + \mathbf{W}_t, \quad \mathbf{W}_t \sim \mathcal{N}_{\text{col}}(\mathbf{0}, \mathbf{\Omega}_t), \quad (2)$$

and the current key reveals one noisy projection of that map,

$$\mathbf{v}_t = \tilde{\mathbf{S}}_t^\top \mathbf{k}_t + \mathbf{e}_t, \quad \mathbf{e}_t \sim \mathcal{N}(\mathbf{0}, r_t \mathbf{I}_{d_v}). \quad (3)$$

Let $\mathcal{F}_t = \sigma\{(\mathbf{k}_\tau, \mathbf{v}_\tau) : \tau \leq t\}$. With a single Gaussian filtering law

$$\tilde{\mathbf{S}}_{t-1} \mid \mathcal{F}_{t-1} \sim \mathcal{N}_{\text{col}}(\mathbf{S}_{t-1}, \mathbf{P}_{t-1}), \quad (4)$$

the Kalman prediction is

$$\hat{\mathbf{S}}_t = \mathbf{D}_t \mathbf{S}_{t-1}, \quad \hat{\mathbf{P}}_t = \mathbf{D}_t \mathbf{P}_{t-1} \mathbf{D}_t^\top + \mathbf{\Omega}_t. \quad (5)$$

Define the innovation, its scalar variance, and the Kalman gain by

$$\boldsymbol{\delta}_t = \mathbf{v}_t - \hat{\mathbf{S}}_t^\top \mathbf{k}_t, \quad \gamma_t = r_t + \mathbf{k}_t^\top \hat{\mathbf{P}}_t \mathbf{k}_t, \quad \boldsymbol{\kappa}_t = \frac{\hat{\mathbf{P}}_t \mathbf{k}_t}{\gamma_t}. \quad (6)$$

Conditioning gives

$$\mathbf{S}_t = \hat{\mathbf{S}}_t + \boldsymbol{\kappa}_t \boldsymbol{\delta}_t^\top, \quad \mathbf{P}_t = (\mathbf{I} - \boldsymbol{\kappa}_t \mathbf{k}_t^\top) \hat{\mathbf{P}}_t. \quad (7)$$

The posterior mean is the minimum-mean-square-error memory, and the layer reads

$$\mathbf{o}_t = \mathbf{S}_t^\top \mathbf{q}_t. \quad (8)$$

The shared-parameter construction below changes only the filtering family. Equations (2) and (3) remain the conditional model in each hypothesis. Component-conditioned dynamics and noise are then introduced as an explicit extension.

3 Gaussian-mixture Kalman Associative Memory

3.1 Mixture filtering law

Replace the single filtering law in Equation (4) by

$$\Pr(Z = c \mid \mathcal{F}_{t-1}) = \pi_{t-1,c}, \quad \tilde{\mathbf{S}}_{t-1} \mid (Z = c, \mathcal{F}_{t-1}) \sim \mathcal{N}_{\text{col}}(\mathbf{S}_{t-1,c}, \mathbf{P}_{t-1,c}), \quad (9)$$

where $\pi_{t-1,c} > 0$ and $\sum_c \pi_{t-1,c} = 1$. Marginally,

$$p(\tilde{\mathbf{S}}_{t-1} \mid \mathcal{F}_{t-1}) = \sum_{c=1}^C \pi_{t-1,c} \mathcal{N}_{\text{col}}(\tilde{\mathbf{S}}_{t-1}; \mathbf{S}_{t-1,c}, \mathbf{P}_{t-1,c}). \quad (10)$$

At $t = 0$, this replaces the original prior $\mathcal{N}_{\text{col}}(\mathbf{0}, \mathbf{I})$ by

$$p(\tilde{\mathbf{S}}_0) = \sum_{c=1}^C \pi_{0,c} \mathcal{N}_{\text{col}}(\tilde{\mathbf{S}}_0; \mathbf{S}_{0,c}, \mathbf{P}_{0,c}). \quad (11)$$

The closest extension of Kalman Associative Memory shares $\mathbf{D}_t$, $\mathbf{\Omega}_t$, and r_t across components. More generally, let the conditional model be

$$\begin{aligned} \tilde{\mathbf{S}}_t \mid Z = c &= \mathbf{D}_{t,c} \tilde{\mathbf{S}}_{t-1} + \mathbf{W}_{t,c}, & \mathbf{W}_{t,c} &\sim \mathcal{N}_{\text{col}}(\mathbf{0}, \mathbf{\Omega}_{t,c}), \\ \mathbf{v}_t \mid (\tilde{\mathbf{S}}_t, Z = c) &= \tilde{\mathbf{S}}_t^\top \mathbf{k}_t + \mathbf{e}_{t,c}, & \mathbf{e}_{t,c} &\sim \mathcal{N}(\mathbf{0}, r_{t,c} \mathbf{I}_{d_v}). \end{aligned} \quad (12)$$

Assume $\mathbf{P}_{0,c} \succ 0$, $\mathbf{\Omega}_{t,c} \succeq 0$, $r_{t,c} > 0$, and that the resulting $\hat{\mathbf{P}}_{t,c}$ is positive definite. Conditioned on Z and the input-dependent quantities $\{(\mathbf{D}_{t,c}, \mathbf{\Omega}_{t,c}, \mathbf{k}_t, r_{t,c})\}$, the process and observation noises have the same temporal and mutual independence assumptions as in Kalman Associative Memory. These quantities are treated as known at filtering time. Tying them across c recovers the original conditional model; the component-indexed notation exposes where meaningful symmetry breaking may enter.

3.2 Exact mixture prediction

Conditioned on $Z = c$, linear-Gaussian prediction gives

$$\hat{\mathbf{S}}_{t,c} = \mathbf{D}_{t,c} \mathbf{S}_{t-1,c}, \quad \hat{\mathbf{P}}_{t,c} = \mathbf{D}_{t,c} \mathbf{P}_{t-1,c} \mathbf{D}_{t,c}^\top + \mathbf{\Omega}_{t,c}. \quad (13)$$

Because Z is persistent, prediction does not move probability between components:

$$\hat{\pi}_{t,c} = \pi_{t-1,c}. \quad (14)$$

Therefore

$$p(\tilde{\mathbf{S}}_t \mid \mathcal{F}_{t-1}) = \sum_{c=1}^C \hat{\pi}_{t,c} \mathcal{N}_{\text{col}}(\tilde{\mathbf{S}}_t; \hat{\mathbf{S}}_{t,c}, \hat{\mathbf{P}}_{t,c}). \quad (15)$$

The number of components remains C : there is one predicted descendant of each persistent hypothesis.

3.3 Component evidence and Bayesian responsibilities

For component c , define

$$\hat{\mathbf{v}}_{t,c} = \hat{\mathbf{S}}_{t,c}^\top \mathbf{k}_t, \quad \boldsymbol{\delta}_{t,c} = \mathbf{v}_t - \hat{\mathbf{v}}_{t,c}, \quad \gamma_{t,c} = r_{t,c} + \mathbf{k}_t^\top \hat{\mathbf{P}}_{t,c} \mathbf{k}_t. \quad (16)$$

Integrating out the component memory gives the predictive observation law

$$\mathbf{v}_t \mid (Z = c, \mathcal{F}_{t-1}, \mathbf{k}_t) \sim \mathcal{N}(\hat{\mathbf{v}}_{t,c}, \gamma_{t,c} \mathbf{I}_{d_v}). \quad (17)$$

Its log evidence is

$$\ell_{t,c} = -\frac{d_v}{2} \log(2\pi\gamma_{t,c}) - \frac{\|\boldsymbol{\delta}_{t,c}\|_2^2}{2\gamma_{t,c}}. \quad (18)$$

Bayes' rule then yields the posterior component probability

$$\pi_{t,c} = \frac{\hat{\pi}_{t,c} \exp(\ell_{t,c})}{\sum_{a=1}^C \hat{\pi}_{t,a} \exp(\ell_{t,a})} = \text{softmax}_c(\log \hat{\pi}_{t,c} + \ell_{t,c}). \quad (19)$$

The variance penalty $d_v \log \gamma_{t,c}$ is part of the Bayesian score. Using only the standardized squared-innovation term $\|\boldsymbol{\delta}_{t,c}\|_2^2 / \gamma_{t,c}$ while omitting this penalty favors diffuse components and is equivalent to the exact likelihood only when all $\gamma_{t,c}$ are equal.

3.4 Conditional Kalman update and exact closure

Every component is conditioned on the observed token. Its gain and posterior parameters are

$$\begin{aligned}\kappa_{t,c} &= \frac{\hat{\mathbf{P}}_{t,c} \mathbf{k}_t}{\gamma_{t,c}}, & \mathbf{S}_{t,c} &= \hat{\mathbf{S}}_{t,c} + \kappa_{t,c} \boldsymbol{\delta}_{t,c}^\top, \\ \mathbf{P}_{t,c} &= (\mathbf{I} - \kappa_{t,c} \mathbf{k}_t^\top) \hat{\mathbf{P}}_{t,c}, & \mathbf{P}_{t,c}^{-1} &= \hat{\mathbf{P}}_{t,c}^{-1} + \frac{1}{r_{t,c}} \mathbf{k}_t \mathbf{k}_t^\top.\end{aligned}\tag{20}$$

The covariance update may equivalently use the symmetric Joseph form

$$\mathbf{P}_{t,c} = (\mathbf{I} - \kappa_{t,c} \mathbf{k}_t^\top) \hat{\mathbf{P}}_{t,c} (\mathbf{I} - \kappa_{t,c} \mathbf{k}_t^\top)^\top + r_{t,c} \kappa_{t,c} \kappa_{t,c}^\top.\tag{21}$$

Proposition 1 (Exact finite-mixture closure). *Suppose Equation (9) holds with one persistent label Z shared across all value columns. Assume the known conditional dynamics and observation model are given by Equation (12), with conditionally independent columns and noises. Then the exact filtering posterior after token t is*

$$p(\tilde{\mathbf{S}}_t \mid \mathcal{F}_t) = \sum_{c=1}^C \pi_{t,c} \mathcal{N}_{\text{col}}(\tilde{\mathbf{S}}_t; \mathbf{S}_{t,c}, \mathbf{P}_{t,c}),\tag{22}$$

where the component parameters are given by Equations (13) and (20), and the weights are given by Equation (19). In particular, exact filtering retains exactly C labeled Gaussian branches at every time.

Proof. Conditioned on $Z = c$, convolution with the linear–Gaussian transition maps the previous columnwise Gaussian to the predicted Gaussian in Equation (13). Multiplying that density by the Gaussian observation likelihood gives the conditional Gaussian in Equation (20); its normalizing constant is precisely $\exp(\ell_{t,c})$. Applying Bayes’ rule to Z normalizes those constants as in Equation (19). Since persistence creates one descendant per component, no additional mixture branches appear. $\square$

Remark 1 (Responsibilities do not attenuate the exact branch update). *The posterior probability $\pi_{t,c}$ changes how much component c contributes to the marginal belief; it does not multiply that component’s Kalman gain. Conditioned on $Z = c$, token t was observed and the branch receives the full update in Equation (20). In particular,*

$$\mathbf{P}_{t,c}^{-1} \neq \hat{\mathbf{P}}_{t,c}^{-1} + \frac{\pi_{t,c}}{r_{t,c}} \mathbf{k}_t \mathbf{k}_t^\top \quad \text{in general}.\tag{23}$$

A responsibility-weighted precision increment can arise from a power-likelihood or mean-field/online-EM approximation to a different model with token-wise assignments and simultaneous memory banks.

4 Bayesian memory, readout, and uncertainty

4.1 Posterior-mean associative memory

Under squared Frobenius loss, the optimal point memory remains the posterior mean,

$$\bar{\mathbf{S}}_t = \mathbb{E}[\tilde{\mathbf{S}}_t \mid \mathcal{F}_t] = \sum_{c=1}^C \pi_{t,c} \mathbf{S}_{t,c}.\tag{24}$$

The Bayesian model-averaged read is therefore

$$\mathbf{o}_t = \bar{\mathbf{S}}_t^\top \mathbf{q}_t = \sum_{c=1}^C \pi_{t,c} \mathbf{S}_{t,c}^\top \mathbf{q}_t. \quad (25)$$

This is a normalized competition among alternative predictions in the same key–value space, not a concatenation of unrelated representation subspaces.

The mean update contains two conceptually different corrections. Define the predicted mixture mean

$$\hat{\mathbf{S}}_t = \sum_{c=1}^C \hat{\pi}_{t,c} \hat{\mathbf{S}}_{t,c}. \quad (26)$$

Substituting Equation (20) into Equation (24) gives

$$\begin{aligned} \bar{\mathbf{S}}_t &= \hat{\mathbf{S}}_t + \sum_{c=1}^C \pi_{t,c} \boldsymbol{\kappa}_{t,c} \boldsymbol{\delta}_{t,c}^\top \\ &\quad + \sum_{c=1}^C (\pi_{t,c} - \hat{\pi}_{t,c}) \hat{\mathbf{S}}_{t,c}. \end{aligned} \quad (27)$$

The second term performs within-hypothesis Kalman corrections. The third moves the marginal mean because the observation changes which hypothesis is credible. Consequently, the model-averaged memory is generally not governed by one ordinary delta rule even though every conditional branch is.

4.2 Within- and between-component uncertainty

Let $\mathbf{m}_{t,c} = \text{vec}(\mathbf{S}_{t,c})$ and $\bar{\mathbf{m}}_t = \text{vec}(\bar{\mathbf{S}}_t)$. The law of total covariance gives

$$\text{Cov}(\text{vec}(\tilde{\mathbf{S}}_t) \mid \mathcal{F}_t) = \sum_{c=1}^C \pi_{t,c} \left[\mathbf{I}_{d_v} \otimes \mathbf{P}_{t,c} + (\mathbf{m}_{t,c} - \bar{\mathbf{m}}_t)(\mathbf{m}_{t,c} - \bar{\mathbf{m}}_t)^\top \right]. \quad (28)$$

The first term is average within-component uncertainty. The second is uncertainty over distinct associative maps. In vectorized memory space, this between-component term has rank at most $C - 1$ and can correlate different key and value coordinates.

Writing $\mathbf{s}_{t,c,j}$ for column j of $\mathbf{S}_{t,c}$ and $\bar{\mathbf{s}}_{t,j}$ for the corresponding mixture mean, the diagonal and off-diagonal value blocks are

$$\begin{aligned} \text{Cov}(\tilde{\mathbf{s}}_{t,j} \mid \mathcal{F}_t) &= \sum_c \pi_{t,c} \left[\mathbf{P}_{t,c} + (\mathbf{s}_{t,c,j} - \bar{\mathbf{s}}_{t,j})(\mathbf{s}_{t,c,j} - \bar{\mathbf{s}}_{t,j})^\top \right], \\ \text{Cov}(\tilde{\mathbf{s}}_{t,j}, \tilde{\mathbf{s}}_{t,\ell} \mid \mathcal{F}_t) &= \sum_c \pi_{t,c} (\mathbf{s}_{t,c,j} - \bar{\mathbf{s}}_{t,j})(\mathbf{s}_{t,c,\ell} - \bar{\mathbf{s}}_{t,\ell})^\top, \quad j \neq \ell. \end{aligned} \quad (29)$$

Thus the marginal mixture is generally not itself $\mathcal{N}_{\text{col}}(\bar{\mathbf{S}}_t, \bar{\mathbf{P}}_t)$: marginalizing the common label induces cross-value dependence even though columns are independent inside each component.

For a query $\mathbf{q}_t$, the uncertainty of the retrieved value is

$$\text{Cov}(\tilde{\mathbf{S}}_t^\top \mathbf{q}_t \mid \mathcal{F}_t) = \sum_c \pi_{t,c} \left[(\mathbf{q}_t^\top \mathbf{P}_{t,c} \mathbf{q}_t) \mathbf{I}_{d_v} + (\mathbf{S}_{t,c}^\top \mathbf{q}_t - \mathbf{o}_t)(\mathbf{S}_{t,c}^\top \mathbf{q}_t - \mathbf{o}_t)^\top \right]. \quad (30)$$

The second term measures disagreement among component reads and has no analogue in a single-Gaussian KDN state.

5 Special cases and the multi-head model

5.1 One component recovers Kalman Associative Memory

Corollary 1 (Single-component boundary). *For $C = 1$, $\pi_{t,1} = 1$, and Equations (13)–(20) reduce exactly to Kalman Associative Memory in Equations (5)–(7).*

This statement concerns the dense optimal filter. To recover Diagonal or Isotropic KDN, impose the same retained covariance family and the same prediction/posterior projection scheme separately in every component; then set $C = 1$. To recover KDA, additionally replace the covariance-derived gain by $\boldsymbol{\kappa}_t = \beta_t \mathbf{k}_t$ and choose $\mathbf{D}_t = \text{diag}(\boldsymbol{\alpha}_t)$. A multi-component fixed-gain model can retain a learned surrogate evidence variance, but without a covariance state its component probabilities are no longer the exact Bayesian probabilities derived above.

5.2 Heads and mixture components are different axes

Assuming independent per-head labels and priors, conditionally independent head processes, and a factorized head observation model, the direct extension to H attention heads is

$$p(\tilde{\mathbf{S}}_{t,1:H} \mid \mathcal{F}_t) = \prod_{h=1}^H \left[\sum_{c=1}^C \pi_{t,h,c} \mathcal{N}_{\text{col}}(\tilde{\mathbf{S}}_{t,h}; \mathbf{S}_{t,h,c}, \mathbf{P}_{t,h,c}) \right]. \quad (31)$$

Every head remains a simultaneously active representation subspace. Components within a head are normalized alternative hypotheses about the same head-local associative map. Therefore the precise reduction statement is:

With the same per-component covariance projection, current multi-head Diagonal or Isotropic KDN is recovered as the single-component ($C = 1$) case of the corresponding per-head Bayesian Gaussian-mixture filter.

It would be incorrect to identify the existing H heads themselves as mixture components. If a single global component label is instead shared across heads, the joint family becomes

$$\sum_{c=1}^C \pi_{t,c} \prod_{h=1}^H \mathcal{N}_{\text{col}}(\tilde{\mathbf{S}}_{t,h}; \mathbf{S}_{t,h,c}, \mathbf{P}_{t,h,c}), \quad (32)$$

and, under conditional independence of head observations given the global label, the component log evidence is summed across heads. That is a distinct, more strongly coupled routing choice.

5.3 When the mixture is nontrivial

If all positive-weight components start with identical means and covariances and use identical dynamics and observation noise, induction through the filter keeps them identical. Their likelihoods remain equal, so the mixture is exactly redundant. A useful construction needs at least one source of symmetry breaking, such as component-specific prior means, prior covariances, process dynamics, process noise, or observation noise. Sharing $\mathbf{k}_t$, $\mathbf{v}_t$, and the underlying coordinate system across components preserves the interpretation of competing hypotheses; giving every component unrelated key and value projections moves the model toward ordinary extra heads or a conventional mixture of experts.

6 Mixture Diagonal KDN

The exact dense Gaussian-sum filter multiplies the dense Riccati state and work by C . A scan-compatible boundary applies the existing Diagonal KDN approximation independently to each component. Let

$$\mathbf{P}_{t-1,c} = \text{diag}(\mathbf{p}_{t-1,c}), \quad \mathbf{D}_{t,c} = \text{diag}(\boldsymbol{\alpha}_{t,c}), \quad \boldsymbol{\Omega}_{t,c} = \text{diag}(\boldsymbol{\omega}_{t,c}). \quad (33)$$

The predictive diagonal covariance is

$$\hat{\mathbf{p}}_{t,c} = \boldsymbol{\alpha}_{t,c}^2 \odot \mathbf{p}_{t-1,c} + \boldsymbol{\omega}_{t,c}. \quad (34)$$

The component innovation variance and gain are

$$\gamma_{t,c} = r_{t,c} + \sum_i \hat{p}_{t,c,i} k_{t,i}^2, \quad \boldsymbol{\kappa}_{t,c} = \frac{\hat{\mathbf{p}}_{t,c} \odot \mathbf{k}_t}{\gamma_{t,c}}. \quad (35)$$

The predictive mean and innovation remain Equations (13) and (16). The component mean is updated by the mean line of Equation (20) using the diagonal gain above. The dense covariance line of that equation is not retained. The component log evidence and posterior weight use Equations (18) and (19). Because their predictive covariance has already been projected at preceding tokens, these are the exact one-step evidence and weight update given the retained diagonal predictive mixture, but approximations to the dense Kalman Associative Memory posterior.

The exact within-component posterior covariance is generally dense. Reverse-KL projection back to the diagonal family preserves the component posterior mean and gives

$$p_{t,c,i} = \left(\hat{p}_{t,c,i}^{-1} + \frac{k_{t,i}^2}{r_{t,c}} \right)^{-1}. \quad (36)$$

Defining $u_{t,c,i} = k_{t,i}^2 / r_{t,c}$ yields the per-component Möbius recurrence

$$p_{t,c,i} = \frac{\alpha_{t,c,i}^2 p_{t-1,c,i} + \omega_{t,c,i}}{u_{t,c,i} \alpha_{t,c,i}^2 p_{t-1,c,i} + 1 + u_{t,c,i} \omega_{t,c,i}}. \quad (37)$$

Equivalently, each (c, i) pair carries the 2×2 map

$$\mathbf{M}_{t,c,i} = \begin{bmatrix} \alpha_{t,c,i}^2 & \omega_{t,c,i} \\ u_{t,c,i} \alpha_{t,c,i}^2 & 1 + u_{t,c,i} \omega_{t,c,i} \end{bmatrix}, \quad (38)$$

whose prefix product, applied to the homogeneous initial vector $[p_{0,c,i}, 1]^\top$, recovers the covariance trajectory through the ratio of its two coordinates. The information scale used by current Diagonal KDN can be introduced by replacing $u_{t,c,i}$ with $\mu_c k_{t,i}^2 / r_{t,c}$. The literal reverse-KL projection is $\mu_c = 1$; other values are calibrated approximations. At a given token, changing the post-write scale does not alter that token's gain or evidence conditional on holding $\hat{\mathbf{p}}_{t,c}$ fixed; it changes later predictive covariances and therefore future gains and evidences.

The mean recurrence for each component is

$$\mathbf{S}_{t,c} = (\mathbf{I} - \boldsymbol{\kappa}_{t,c} \mathbf{k}_t^\top) \mathbf{D}_{t,c} \mathbf{S}_{t-1,c} + \boldsymbol{\kappa}_{t,c} \mathbf{v}_t^\top. \quad (39)$$

Once $\boldsymbol{\kappa}_{t,c}$ is obtained from the covariance scan, this is the same affine two-key memory recurrence already used by Diagonal KDN.

7 Associative structure and parallel schedule

The persistent mixture has an important factorization: neither the component covariance update nor the component mean update depends on $\pi_{t,c}$. Posterior weights couple component *beliefs*, but they do not feed back into conditional component trajectories. Consequently, the C conditional filters can be evaluated independently before their probabilities and outputs are combined.

Let $a_{t,c}$ denote an unnormalized log weight. Since $\hat{\pi}_{t,c} = \pi_{t-1,c}$,

$$a_{t,c} = \log \pi_{0,c} + \sum_{\tau=1}^t \ell_{\tau,c}, \quad \pi_{t,c} = \exp(a_{t,c} - \text{logsumexp}_b a_{t,b}). \quad (40)$$

Thus mixture routing itself is an additive prefix scan once the component evidences are known.

Proposition 2 (Staged scan factorization). *For persistent Mixture Diagonal KDN, the defined projected recurrence can be evaluated exactly in the following order:*

1. run C independent Möbius covariance scans to obtain $\hat{\mathbf{p}}_{t,c}$ and $\boldsymbol{\kappa}_{t,c}$;
2. run C independent affine memory scans to obtain component reads and exclusive innovations $\boldsymbol{\delta}_{t,c}$;
3. form $\ell_{t,c}$, perform the additive prefix scans in Equation (40), and normalize across c ;
4. reduce component reads using Equation (25).

No component-state recurrence depends on the result of the weight scan.

Proof. Equations (34)–(39) contain only component-local state and token inputs. They therefore determine every component trajectory without reference to $\pi_{t,c}$. Equation (18) is then a tokenwise function of the exclusive component trajectory, and persistent posterior odds are the prefix sum in Equation (40). The only cross-component operations are the token-local log-sum-exp normalization and weighted output reduction. $\square$

This factorization corrects a tempting but overly broad conclusion: state-dependent Bayesian responsibilities do not destroy the existing scan structure in the persistent model, because they do not control the branch updates. The component dimension can in principle be flattened into the head dimension for the existing gain and memory algebra, but the current public kernels need interface changes. A naive flatten physically replicates shared keys, values, and queries; a native component axis or stride-aware loads would avoid that bandwidth cost. The layer frontend, normalization, and output projection also retain the original head count. The key new kernel contract is access to the exclusive innovation and innovation variance for every token and component. Reconstructing the innovation from a post-write read requires division by $1 - \mathbf{k}_t^\top \boldsymbol{\kappa}_{t,c}$ and becomes unstable near a full write, so a production kernel should expose an exclusive key read or fuse a dedicated innovation computation and its backward.

For H heads and C components per head, the retained states scale as

branch family	state per layer, up to batch factors
Isotropic KDN mixture	$O(HC(d_k d_v + 1))$
Diagonal KDN mixture	$O(HC(d_k d_v + d_k + 1))$
Dense Kalman mixture	$O(HC(d_k d_v + d_k^2 + 1))$.

(41)

These are recurrent inference states; the final scalar accounts for each component’s log weight. For distinct component means, all components process all tokens, so the persistent Gaussian-sum model

offers no sparse-compute benefit. Assuming diagonal or similarly structured transitions, a diagonal implementation requires $O(BTHC(d_k d_v + d_k))$ branch work, followed by $O(BTHC d_v + BTHC)$ evidence, normalization, and reduction work. Dense recurrent filtering costs $O(BTHC(d_k d_v + d_k^2))$ under the same transition structure; a generic dense transition instead adds $O(d_k^2 d_v + d_k^3)$ work per component and token. If $\mathbf{P}_{0,c}, \mathbf{D}_{t,c}, \mathbf{\Omega}_{t,c}, r_{t,c}$ are tied, covariance and gain trajectories are identical and that scan can be shared; distinct means still require component-specific memory work. If the means are tied as well, the mixture is redundant.

Materializing $[B, T, H, C, d_v]$ innovations and component reads can be expensive. Innovation norms can be reduced immediately after evidence formation, although backward must retain or recompute the needed residual information. Component reads cannot be reduced until the prefix evidence scan produces posterior weights, so they must be retained, recomputed, or handled by a more involved chunk pipeline.

8 A unified latent routed-memory model

The persistent Gaussian sum is an exact anchor, but its component label describes a sequence-global hypothesis rather than scalable token routing. We now enlarge the model so that the categorical assignment can change with the token. This gives one probabilistic family whose persistent boundary is the exact filter above and whose variational approximation is a routed bank of associative memories.

8.1 A conditional mixture of dynamic associative memories

Maintain C simultaneous latent memories $\tilde{\mathbf{S}}_{t,1:C}$. Their joint transition is

$$p(\tilde{\mathbf{S}}_{t,1:C} \mid \tilde{\mathbf{S}}_{t-1,1:C}, \mathbf{x}_t) = \prod_{c=1}^C \mathcal{N}_{\text{col}}\left(\tilde{\mathbf{S}}_{t,c}; \mathbf{D}_{t,c} \tilde{\mathbf{S}}_{t-1,c}, \mathbf{\Omega}_{t,c}\right), \quad (42)$$

where the process noises are conditionally independent. Given the observed token representation $\mathbf{x}_t$, draw a local assignment

$$Z_t \mid \mathbf{x}_t \sim \text{Cat}(\boldsymbol{\pi}_{\theta,t}), \quad \boldsymbol{\pi}_{\theta,t} = \text{softmax}(\mathbf{g}_{\theta}(\mathbf{x}_t)), \quad (43)$$

and let only the selected memory explain the current key-value observation,

$$\tilde{\mathbf{S}}_t^{\text{act}} := \tilde{\mathbf{S}}_{t,Z_t}, \quad \mathbf{v}_t \mid (Z_t = c, \tilde{\mathbf{S}}_{t,c}, \mathbf{k}_t) \sim \mathcal{N}(\tilde{\mathbf{S}}_{t,c}^{\top} \mathbf{k}_t, r_{t,c} \mathbf{I}_{d_v}). \quad (44)$$

Thus Kalman Associative Memory is dynamic Bayesian linear regression from keys to values, while the routed model is a conditional mixture of dynamic Bayesian linear regressions.

More generally, a route transition

$$T_{t,ac} = \Pr(Z_t = c \mid Z_{t-1} = a, \mathbf{x}_t) \quad (45)$$

unifies several cases. The identity transition $T_{t,ac} = \mathbf{1}\{a = c\}$ gives a persistent route. Identical rows $T_{t,ac} = \pi_{\theta,t,c}$ give a fresh token-local route, and a diagonally biased matrix gives a sticky switching model. The persistent case is exactly closed because each component has one descendant. General switching is not.

To see the connection to the earlier Gaussian sum, use the identity route transition, set $\Pr(Z_0 = c) = \pi_{0,c}$, initialize bank c with the c th component prior, and let all banks follow Equation (42). Conditional on $Z = c$, only bank c supplies observations. Integrating out the unused banks gives exactly the marginal law of the active state $\tilde{\mathbf{S}}_t^{\text{act}}$ in Section 3. Thus the earlier one-state mixture is a marginal boundary of this simultaneous-bank construction.

8.2 Fresh-route one-step posterior and route-history growth

Let

$$\hat{q}_{t,c}(\tilde{\mathbf{S}}_{t,c}) = \mathcal{N}_{\text{col}}(\tilde{\mathbf{S}}_{t,c}; \hat{\mathbf{S}}_{t,c}, \hat{\mathbf{P}}_{t,c}) \quad (46)$$

denote the predicted Gaussian factor for bank c . In the fresh-route case, start from the factorized predictive law

$$\tilde{p}_t(Z_t, \tilde{\mathbf{S}}_{t,1:C}) = \text{Cat}(Z_t; \boldsymbol{\pi}_{\theta,t}) \prod_{c=1}^C \hat{q}_{t,c}(\tilde{\mathbf{S}}_{t,c}). \quad (47)$$

This law is exact at the first token under independent bank priors, and after later tokens it is the fixed-size product projection used below. Conditioning this law on one observation gives the exact one-step posterior

$$\begin{aligned} \tilde{p}_t(Z_t = c, \tilde{\mathbf{S}}_{t,1:C} \mid \mathbf{v}_t, \mathbf{k}_t, \mathbf{x}_t) \\ = \rho_{t,c}^* q_{t,c}^+(\tilde{\mathbf{S}}_{t,c}) \prod_{b \neq c} \hat{q}_{t,b}(\tilde{\mathbf{S}}_{t,b}), \end{aligned} \quad (48)$$

where $q_{t,c}^+$ is the full Kalman update of bank c and

$$\rho_{t,c}^* \propto \pi_{\theta,t,c} \mathcal{N}(\mathbf{v}_t; \hat{\mathbf{S}}_{t,c}^\top \mathbf{k}_t, \gamma_{t,c} \mathbf{I}_{d_v}). \quad (49)$$

This is already a mixture of C joint bank distributions. At the next token, without the product projection, each route history generally produces different bank states and can branch again. Exact filtering can therefore grow as C^t . A Markov route has the same continuous-state problem: although its discrete weights admit standard semiring operations, every destination receives a mixture over predecessor bank states. For a Markov route, replacing $\boldsymbol{\pi}_{\theta,t}$ by a predicted route marginal in Equation (47) is itself part of the projection, not an exact filtering identity. A fixed-size implementation needs an explicit projection or merging rule.

When $Z_t = Z$ is persistent, only the branch that repeatedly selects c matters under hypothesis $Z = c$; marginalizing unused banks recovers exactly the C -component Gaussian-sum filter in Section 3. The exact persistent model is therefore a boundary of the routed-bank model, not the sparse implementation itself.

9 Amortized online variational filtering

9.1 Structured mean-field projection

After every token, project the route-history posterior onto

$$q_t(Z_t, \tilde{\mathbf{S}}_{t,1:C}) = \text{Cat}(Z_t; \boldsymbol{\rho}_t) \prod_{c=1}^C q_{t,c}(\tilde{\mathbf{S}}_{t,c}), \quad q_{t,c} = \mathcal{N}_{\text{col}}(\mathbf{S}_{t,c}, \mathbf{P}_{t,c}). \quad (50)$$

For predicted bank factors $\hat{q}_{t,c}$, define the one-step objective relative to the projected predictive law in Equation (47):

$$\mathcal{L}_t(q) = \mathbb{E}_q[\log p(\mathbf{v}_t \mid Z_t, \tilde{\mathbf{S}}_{t,1:C}, \mathbf{k}_t)] - \text{KL}(q_t \parallel \tilde{p}_t). \quad (51)$$

Using the factorization in Equation (50), this expands to

$$\begin{aligned} \mathcal{L}_t(q) = & \sum_{c=1}^C \rho_{t,c} \mathbb{E}_{q_{t,c}} \left[\log p(\mathbf{v}_t \mid \tilde{\mathbf{S}}_{t,c}, Z_t = c, \mathbf{k}_t) \right] \\ & - \text{KL}(\text{Cat}(\boldsymbol{\rho}_t) \parallel \text{Cat}(\boldsymbol{\pi}_{\theta,t})) - \sum_{c=1}^C \text{KL}(q_{t,c} \parallel \hat{q}_{t,c}). \end{aligned} \quad (52)$$

For fixed $\boldsymbol{\rho}_t$, the coordinate-optimal bank factor is

$$q_{t,c}(\tilde{\mathbf{S}}_{t,c}) \propto \hat{q}_{t,c}(\tilde{\mathbf{S}}_{t,c}) p(\mathbf{v}_t \mid \tilde{\mathbf{S}}_{t,c}, Z_t = c, \mathbf{k}_t)^{\rho_{t,c}}. \quad (53)$$

Thus the local responsibility allocates the token’s observation precision across the memory bank. With $\boldsymbol{\Lambda}_{t,c} = \mathbf{P}_{t,c}^{-1}$ and $\boldsymbol{\Xi}_{t,c} = \boldsymbol{\Lambda}_{t,c} \mathbf{S}_{t,c}$, the update is

$$\begin{aligned} \boldsymbol{\Lambda}_{t,c} &= \hat{\mathbf{P}}_{t,c}^{-1} + \frac{\rho_{t,c}}{r_{t,c}} \mathbf{k}_t \mathbf{k}_t^\top, \\ \boldsymbol{\Xi}_{t,c} &= \hat{\mathbf{P}}_{t,c}^{-1} \hat{\mathbf{S}}_{t,c} + \frac{\rho_{t,c}}{r_{t,c}} \mathbf{k}_t \mathbf{v}_t^\top. \end{aligned} \quad (54)$$

Equivalently,

$$\begin{aligned} \mathbf{S}_{t,c} &= \hat{\mathbf{S}}_{t,c} + \boldsymbol{\kappa}_{t,c}^{(\rho)} \left(\mathbf{v}_t - \hat{\mathbf{S}}_{t,c}^\top \mathbf{k}_t \right)^\top, \\ \boldsymbol{\kappa}_{t,c}^{(\rho)} &= \frac{\rho_{t,c} \hat{\mathbf{P}}_{t,c} \mathbf{k}_t}{r_{t,c} + \rho_{t,c} \mathbf{k}_t^\top \hat{\mathbf{P}}_{t,c} \mathbf{k}_t}. \end{aligned} \quad (55)$$

Derivation of the bank coordinate. Holding $\boldsymbol{\rho}_t$ and the other bank factors fixed, the terms in Equation (52) that depend on $q_{t,c}$ give

$$\log q_{t,c}^*(\tilde{\mathbf{S}}_{t,c}) = \log \hat{q}_{t,c}(\tilde{\mathbf{S}}_{t,c}) + \rho_{t,c} \log p(\mathbf{v}_t \mid \tilde{\mathbf{S}}_{t,c}, Z_t = c, \mathbf{k}_t) + \text{const}. \quad (56)$$

The first term is Gaussian and the second is a Gaussian log likelihood multiplied by $\rho_{t,c}$. Collecting the quadratic and linear terms in $\tilde{\mathbf{S}}_{t,c}$ gives the precision and precision-weighted mean in Equation (54). Applying the matrix inversion lemma gives Equation (55). $\square$

When $\rho_{t,c} = 0$, bank c receives no measurement information; when $\rho_{t,c} = 1$, it receives the ordinary Kalman update. These are exact coordinate updates for Equation (52) given fixed responsibilities, but the factorized family is an approximation to exact route-history filtering.

The likelihood power in Equation (53) is not an ad hoc temperature. It is the ordinary mean-field coordinate update: the selector $\mathbf{1}\{Z_t = c\}$ appears in the complete-data log likelihood, and its expectation under the categorical factor is $\rho_{t,c}$. If $\rho_{t,c}$ is instead supplied by a deterministic gate trained only through a language-model objective, the same recurrence is better viewed as a generalized-Bayes or stateful-MoE update motivated by this coordinate calculation.

This is the direct dynamic analogue of finite-mixture variational inference [1]: a static component parameter becomes a recurrent Gaussian associative-memory bank, the local assignment becomes Z_t , its responsibility becomes $\rho_{t,c}$, and responsibility-weighted sufficient statistics become the fractional precision and precision-weighted-mean increments in Equation (54).

9.2 Exact and amortized responsibilities

The unconstrained categorical coordinate satisfies

$$\rho_{t,c}^{\text{CAVI}} \propto \pi_{\theta,t,c} \exp \left\{ \mathbb{E}_{q_{t,c}} [\log p(\mathbf{v}_t \mid \tilde{\mathbf{S}}_{t,c}, Z_t = c, \mathbf{k}_t)] \right\}. \quad (57)$$

For a Gaussian bank factor, its expected log likelihood is

$$\begin{aligned} & \mathbb{E}_{q_{t,c}} [\log p(\mathbf{v}_t \mid \tilde{\mathbf{S}}_{t,c}, Z_t = c, \mathbf{k}_t)] \\ &= -\frac{d_v}{2} \log(2\pi r_{t,c}) - \frac{\|\mathbf{v}_t - \mathbf{S}_{t,c}^\top \mathbf{k}_t\|_2^2 + d_v \mathbf{k}_t^\top \mathbf{P}_{t,c} \mathbf{k}_t}{2r_{t,c}}. \end{aligned} \quad (58)$$

This coordinate depends on recurrent bank state, so alternating it with Equation (54) is sequential and would obstruct the existing input-coefficient scan.

The scalable restriction is an amortized local factor

$$q_\phi(Z_t = c \mid \mathbf{x}_t) = \rho_{t,c}, \quad \boldsymbol{\rho}_t = \text{softmax}(\mathbf{g}_\phi(\mathbf{x}_t)), \quad (59)$$

computed from pre-mixer token features only. It approximates the state-aware categorical coordinate but is not the exact Bayesian responsibility in Equation (49). Keeping the generative prior $\pi_{\theta,t}$ distinct from the recognition probability $\boldsymbol{\rho}_t$ preserves the KL term in Equation (52). Tying them is a simpler conditional-gating model, but weakens the inference interpretation. If training uses only language-model loss and no variational or calibration objective, the most accurate claim is that the router and updates are *variationally derived*, not that the layer performs optimized variational inference.

Using the same local factor for reading gives

$$\mathbf{o}_t = \sum_{c=1}^C \rho_{t,c} \mathbf{S}_{t,c}^\top \mathbf{q}_t. \quad (60)$$

Separate read and write gates are possible, but Equation (60) is the coherent boundary in which one latent assignment controls both. Components share key, value, and query projections inside each original head, compete on a simplex, and reduce back to the original head width before the output projection. This distinguishes the factorization from simply widening the model with unrelated heads, although it remains representable as a constrained set of tied, gated heads.

In the recommended shared-across-head variant, one global Z_t selects the same component index jointly across all H original heads. The headwise expected log likelihoods add in Equation (52), and the same $\rho_{t,c}$ weights every head. Independent variables $Z_{t,h}$ instead define a per-head routing family with different semantics and many more routed streams. In neither case do the original representation heads become mixture components. Formally, the shared-route factorization uses

$$q_t = \text{Cat}(Z_t; \boldsymbol{\rho}_t) \prod_{h=1}^H \prod_{c=1}^C q_{t,h,c}, \quad p(\mathbf{v}_{t,1:H} \mid Z_t = c, \tilde{\mathbf{S}}_{t,1:H,c}) = \prod_{h=1}^H p(\mathbf{v}_{t,h} \mid Z_t = c, \tilde{\mathbf{S}}_{t,h,c}). \quad (61)$$

9.3 Routed Diagonal KDN remains scan-compatible

For a diagonal bank covariance, define

$$\begin{aligned}\hat{\mathbf{p}}_{t,c} &= \boldsymbol{\alpha}_{t,c}^2 \odot \mathbf{p}_{t-1,c} + \boldsymbol{\omega}_{t,c}, \\ \boldsymbol{\kappa}_{t,c}^{(\rho)} &= \frac{\rho_{t,c} \hat{\mathbf{p}}_{t,c} \odot \mathbf{k}_t}{r_{t,c} + \rho_{t,c} \sum_i \hat{p}_{t,c,i} k_{t,i}^2}, \quad \mathbf{u}_{t,c} = \frac{\mu_c \rho_{t,c}}{r_{t,c}} \mathbf{k}_t \odot \mathbf{k}_t, \\ \mathbf{p}_{t,c} &= \frac{\hat{\mathbf{p}}_{t,c}}{1 + \mathbf{u}_{t,c} \odot \hat{\mathbf{p}}_{t,c}}.\end{aligned}\tag{62}$$

The literal variational update has $\mu_c = 1$; other values retain the existing KDN information calibration. Substituting $u_{t,c,i} = \mu_c \rho_{t,c} k_{t,i}^2 / r_{t,c}$ into Equation (38) gives the same 2×2 Möbius scan for every bank and key channel. The mean update remains

$$\mathbf{S}_{t,c} = (\mathbf{I} - \boldsymbol{\kappa}_{t,c}^{(\rho)} \mathbf{k}_t^\top) \mathbf{D}_{t,c} \mathbf{S}_{t-1,c} + \boldsymbol{\kappa}_{t,c}^{(\rho)} \mathbf{v}_t^\top.\tag{63}$$

Because Equation (59) makes every $\rho_{t,c}$ known before either recurrence, the covariance maps and affine memory maps remain associative. A router that reads the current bank predictions would reintroduce a state-routing cycle.

Corollary 2 (Routed single-component boundary). *For $C = 1$, $\rho_{t,1} = 1$. Equations (62) and (63) reduce to current Diagonal KDN with the same information-scale choice. Replacing the covariance-derived gain by the fixed gain $\beta_t \mathbf{k}_t$ gives the corresponding KDA boundary.*

For a routed fixed-gain KDA bank with $C > 1$, the direct control baseline uses

$$\boldsymbol{\kappa}_{t,c}^{\text{KDA}} = \rho_{t,c} \beta_{t,c} \mathbf{k}_t.\tag{64}$$

This is not a covariance-derived variational gain, but it preserves the same input-known affine scan and gives an inexpensive systems prototype for the routing mechanism.

10 Sparse capacity scaling

10.1 Dense soft routing is the reference recurrence

With a dense softmax, every $\rho_{t,c}$ is nonzero, so all C memories must update and read every token. This gives the cleanest reference implementation and directly validates the variational recurrence, but its recurrent work and training activations are C times the one-memory layer. It scales representational state, not computational efficiency.

Merely setting most variational responsibilities to zero does not by itself freeze the associated latent dynamics. For a genuinely sparse local-clock model, add a deterministic participation controller $\mathcal{A}_t$ of size $k \ll C$ to the conditional generative model. Define

$$\begin{aligned}\mathcal{A}_t &= \text{TopK}(\mathbf{g}_\psi(\mathbf{x}_t), k), & a_{t,c} &= \mathbf{1}\{c \in \mathcal{A}_t\}, \\ \mathbf{D}_{t,c}^{\mathcal{A}} &= a_{t,c} \mathbf{D}_{t,c} + (1 - a_{t,c}) \mathbf{I}, & \boldsymbol{\Omega}_{t,c}^{\mathcal{A}} &= a_{t,c} \boldsymbol{\Omega}_{t,c}, \\ \pi_{\theta,t,c} &= \rho_{t,c} = 0 \quad (c \notin \mathcal{A}_t), & \sum_{c \in \mathcal{A}_t} \pi_{\theta,t,c} &= \sum_{c \in \mathcal{A}_t} \rho_{t,c} = 1.\end{aligned}\tag{65}$$

The quantities $\mathbf{D}^{\mathcal{A}}$ and $\mathbf{\Omega}^{\mathcal{A}}$ summarize the conditional transition moments. Literally, the joint bank transition in Equation (42) is replaced by the product of kernels

$$p(\tilde{\mathbf{S}}_{t,1:C} \mid \tilde{\mathbf{S}}_{t-1,1:C}, \mathbf{x}_t, \mathcal{A}_t) = \prod_{c=1}^C \kappa_{t,c}^{\mathcal{A}}(\tilde{\mathbf{S}}_{t,c}), \quad (66)$$

$$\kappa_{t,c}^{\mathcal{A}}(\tilde{\mathbf{S}}) = \begin{cases} \mathcal{N}_{\text{col}}(\tilde{\mathbf{S}}; \mathbf{D}_{t,c} \tilde{\mathbf{S}}_{t-1,c}, \mathbf{\Omega}_{t,c}), & a_{t,c} = 1, \\ \delta_{\tilde{\mathbf{S}}_{t-1,c}}(\tilde{\mathbf{S}}), & a_{t,c} = 0. \end{cases}$$

Here δ is a Dirac measure; after integrating over the previous bank factor, an inactive bank’s predictive factor is simply unchanged. The route prior and recognition factor are both supported inside $\mathcal{A}_t$; they need not otherwise be equal. Equations (47)–(52) apply conditional on $\mathcal{A}_t$, using predictive factors generated by this kernel and masked categorical factors. Event-time freezing is therefore a declared model choice, not an implication of top- k posterior truncation.

Top-2 is the recommended initial point because it preserves competition and ordinary gradients between the two selected recognition scores while bounding recurrent work. A first implementation can tie $\mathbf{g}_\psi = \mathbf{g}_\phi$ and apply a masked softmax. Hard set membership and unselected scores still have no pathwise gradient; a distinct participation network needs a straight-through estimator, sampling relaxation, or auxiliary routing objective. Top-1 is a useful ablation but also needs such a mechanism because renormalizing its only selected probability to one removes the within-route gate gradient. In the exact categorical model, one $Z_t \in \mathcal{A}_t$ supplies the likelihood and all participating banks undergo their declared dynamics. In the projected filter, every bank with $\rho_{t,c} > 0$ additionally receives its fractional measurement update. Consequently a runner-up with tiny $\rho_{t,c}$ still undergoes a full transition but receives almost no measurement; gating the transition itself by responsibility would define another model and should be ablated separately.

10.2 Local event time: the sparse reference semantics

Under Equation (66), an inactive memory is exactly frozen. For each bank, let $t_{c,1} < \dots < t_{c,N_c}$ be the tokens at which $c \in \mathcal{A}_t$ and use a local event index j :

$$\begin{aligned} \tilde{\mathbf{S}}_{j,c} &= \mathbf{D}_{t_{c,j},c} \tilde{\mathbf{S}}_{j-1,c} + \mathbf{W}_{t_{c,j},c}, \\ q_{j,c}(\tilde{\mathbf{S}}_{j,c}) &\propto \hat{q}_{j,c}(\tilde{\mathbf{S}}_{j,c}) p(\mathbf{v}_{t_{c,j}} \mid \tilde{\mathbf{S}}_{j,c}, Z_{t_{c,j}} = c, \mathbf{k}_{t_{c,j}})^{\rho_{t_{c,j},c}}. \end{aligned} \quad (67)$$

Between routed events, the bank transition is the identity and process noise is zero. Dropping inactive tokens from that bank’s stream is therefore exact. Stable dispatch by (batch, c, t) produces chronological variable-length sequences, each of which can be processed by the ordinary associative recurrence. Outputs are inverse-scattered to token order and combined by Equation (60).

This event-time model deliberately differs from global-time KDN: a rarely used memory does not decay merely because unrelated tokens were routed elsewhere. That behavior is natural for a memory-slot interpretation and gives the cleanest $O(k)$ recurrent computation, but it must be stated as part of the model rather than hidden as a kernel shortcut.

10.3 Global-time aging as a controlled alternative

If every memory must age at every global token, use the original all-bank transition in Equation (42) and sparsify only the measurement update. A bank outside the measurement active set still performs

$$\mathbf{S}_{t,c} = \mathbf{D}_{t,c} \mathbf{S}_{t-1,c}, \quad \mathbf{p}_{t,c} = \boldsymbol{\alpha}_{t,c}^2 \odot \mathbf{p}_{t-1,c} + \boldsymbol{\omega}_{t,c}. \quad (68)$$

Sparse execution is still possible when skipped transitions have compact interval summaries. A diagonal mean transition composes by products of α , while each covariance coordinate has the affine map $p \mapsto ap + b$; these maps compose associatively over the gap between routed events. If transition parameters are shared across memories, their prefix summaries can be computed once and queried at routed events. Fully component- and token-specific transitions generally require $O(CT)$ transition generation or a new hierarchical parameterization, weakening the scaling benefit.

For KDA, a skipped interval can use differences of prefix sums of the activated full-timeline log-decay, then call the scan with that already activated gate (`use_gate_in_kernel=False`). For Diagonal KDN, the skipped prediction is the composed affine map $p \mapsto ap + b$, followed at the next selected event by its Möbius measurement map. A global-clock implementation must therefore retain a timestamp or pending interval transform for every inactive bank during streaming or stateful execution; an offline full-sequence implementation can instead materialize the relevant prefix differences. In either case it cannot interpret an unchanged stored state as already aged.

10.4 Segmented scan and complexity

For a router shared across the original heads, one global assignment selects the same component index jointly across all H heads. Stable routing then creates at most BC bank streams, each retaining the original H heads, and a Bayesian or CAVI score sums its headwise evidence. A per-head route creates up to BHC streams and substantially more metadata and short segments, so shared-across-head routing is the preferred first systems boundary. Keys, values, and queries remain shared across components.

After routing and compaction, the sparse layer performs:

1. compute router logits and a top- k active set from pre-mixer token states;
2. stable-sort routed events by bank while preserving time within each stream;
3. run a variable-length segmented KDA/KDN scan on nonempty streams;
4. inverse-scatter component reads and reduce their weighted sum;
5. scatter final states back through the active-stream-to-bank map. Under the local clock, omitted banks remain untouched; under the global clock, retain and apply their pending interval transform and final tail.

Chronological stability is a correctness condition: unordered atomic dispatch changes the recurrent model. Omit zero-length streams from packed `cu.seqlens`; retain an explicit active-stream-to-bank map for the final scatter. Passing repeated offsets through metadata builders that assume positive lengths is not a safe representation of empty banks.

For Diagonal KDN, dense soft routing costs

$$\Theta(BTHC(d_k d_v + d_k)) \tag{69}$$

in branch work. Ideal top- k event-time routing reduces this to

$$\Theta(BTHk(d_k d_v + d_k)) + \Theta(BTd_{\text{model}}C) \tag{70}$$

for ideal recurrent branch work plus dense linear router scoring shared across heads. The router also materializes $O(BTC)$ logits. State storage remains $\Theta(BHC(d_k d_v + d_k))$; sparsity scales recurrent-branch compute and activations, not router work, dispatch metadata, or the persistent cache. For very large C , dense router logits eventually require hierarchical, product-key, or approximate top- k selection. Sorting, prefix construction, gather/scatter, and chunk padding add costs not shown in

Equation (70); when the mean stream length Tk/C is much smaller than a hardware-efficient chunk, those costs can dominate.

The nearest implementation substrate is `lit_gpt/kda_ops/chunk.py::chunk_kda`. Its variable-length interface uses a flattened $B = 1$ token buffer, `cu_seqlens`, and per-stream FP32 initial and final states of shape $[N, H, K, V]$. Its current chunk-index preparation calls `tolist()`, so fully dynamic routing ultimately needs GPU-native histogram, prefix-offset, and chunk-index construction to avoid a host synchronization. Short streams also waste partial 64-token chunks.

Sparse Diagonal KDN additionally needs ragged initial and final covariance state and a ragged asymmetric-WY memory scan. The current `kla_kappa_chunk` does not support `initial_info`, and the public `chunk_kalman` wrapper has no variable-length interface. The recommended sequence is therefore a dense-soft PyTorch reference, followed by a shared-across-head top-2 KDA bank, followed by sparse Diagonal KDN only after the routing hypothesis earns the additional kernel work.

11 Parameterization and failure modes

Gating prior and recognition router. The conditional prior $\pi_{\theta,t}$ and variational responsibility ρ_t play different roles. In the dense model, a simple first implementation can use a uniform prior and one learned $q_\phi(Z_t | \mathbf{x}_t)$. In that case $\text{KL}(\text{Cat}(\rho_t) \| \text{Cat}(\mathbf{1}/C)) = \log C - \mathcal{H}(\rho_t)$ is a *per-token entropy* regularizer, not a load-balancing term. In the sparse model, a prior uniform on $\mathcal{A}_t$ instead gives $\log k - \mathcal{H}(\rho_t)$. Balance across a batch or sequence requires a separate aggregate statistic, for example $\bar{\rho}_c = (BT)^{-1} \sum_{b,t} \rho_{bt,c}$ and $\text{KL}(\text{Cat}(\bar{\rho}) \| \text{Cat}(\mathbf{1}/C))$, or an explicit hard-dispatch load penalty. A more literal conditional latent-variable model learns both networks and trains the local ELBO alongside the language-model objective. If no such objective is used, the probabilistic derivation still specifies the update but should be described as motivation rather than optimized inference.

Shared observation space. Use common key and value projections within a head so components explain the same observation. Component-specific transition, process-noise, observation-noise, or prior parameters provide controlled hypothesis diversity without turning components into unrelated heads.

Symmetry, collapse, and balance. The likelihood is invariant to a permutation of component labels. Identical memories form an exact fixed point only when the router and update preserve that symmetry; deterministic top- k tie-breaking can break it immediately. Initialization should therefore be deliberately weak and component-permutation equivariance should be tested away from tied routing logits, or with an explicitly equivariant tie rule. A token-local router can also starve memories before they specialize. Report token fractions, probability mass, effective component count, and maximum load per component; use an aggregate balancing loss only as a declared modeling choice.

Persistent and amortized collapse differ. For persistent Z , log odds accumulate evidence over the whole sequence. A component that loses early can acquire exponentially small posterior weight, and under identifiable, persistently informative data the filter can concentrate toward one sequence-global hypothesis. This is coherent Bayesian behavior but may not provide span-level routing. In the amortized bank, collapse instead means that the recognition network repeatedly selects a few components. Weight forgetting, floors, entropy penalties, aggregate load penalties, component resets, or likelihood tempering can mitigate either failure, but each changes the probabilistic model or

objective and must be identified explicitly. Top- k alone is only a compute constraint and does not prevent collapse onto the same small set of banks.

High-dimensional evidence. Both terms in Equation (18) scale with d_v . Learned value coordinates are neither guaranteed independent nor perfectly Gaussian, so the nominal likelihood may become extremely sharp. Dividing the log evidence by d_v or introducing a temperature can calibrate routing, but the exact Bayesian baseline is the untempered expression. Log weights, quadratic residual accumulation, and log-sum-exp normalization should use at least FP32. The input-only amortized router avoids this numerical path in its forward decision, but a state-aware teacher or auxiliary variational objective still inherits the calibration issue.

Covariance scale. As in ordinary Kalman Associative Memory, jointly scaling predictive covariance and observation noise by a positive factor leaves the gain unchanged. It does not leave mixture evidence unchanged. A common factor rescales every innovation variance and therefore tempers the relative quadratic evidence, even though its common log-determinant offset cancels across components. Component-specific factors additionally change determinant comparisons. Mixture responsibilities therefore expose covariance-calibration errors that can be invisible in the mean update alone.

12 Validation contract

Before a language-model experiment, a reference implementation should pass:

1. **Single-component reduction.** At $C = 1$, match the dense KAM, Diagonal KDN, or Isotropic KDN reference selected for each test, including forward, backward, and final state.
2. **Persistent exact anchor.** For small d_k, d_v, C, T , compare component means, covariances, evidences, and weights with direct Gaussian density evaluation.
3. **Mixture-moment check.** Compare Equations (24) and (28) with Monte Carlo samples from the filtered mixture.
4. **Fractional-update identity.** In FP64, verify that the natural-parameter and gain forms agree for $\rho \in \{0, 10^{-6}, 0.1, 0.5, 1\}$.
5. **Dense routed scan parity.** For fixed token-local responsibilities, compare every Möbius scan, gain, component memory, and routed read with a sequential FP64 oracle.
6. **Router invariants.** Check simplex normalization, top- k support, causal dependence on pre-mixer inputs only, and finite gradients under nearly tied and extremely separated logits.
7. **Permutation equivariance.** Away from tied logits, jointly permuting component parameters, router logits, and states must permute internals while leaving the reduced output unchanged. Test ties separately only with an equivariant tie rule.
8. **Sparse dispatch parity.** Compare packed event-time execution with a dense implementation of Equation (66) under empty banks, highly imbalanced loads, partial chunks, and repeated routing. Verify chronological order within every bank stream.
9. **Route-history audit.** At tiny C and T , enumerate exact token routes and compare the variational approximation’s means, covariances, and predictive distribution with the true mixture rather than treating the fractional update as exact Bayes.

10. **Clock semantics.** If global-time aging is implemented, compare its lazy interval maps with a dense all-bank recurrence; never compare it against the event-time model as if they were the same recurrence.

The decisive first experiment should separate routed memory capacity from increased state, parameters, and compute. At small scale, compare $C = 1$, a dense-soft $C > 1$ reference, a top-2 event-time bank, an expanded-head control with comparable state, a random or hashed router, and an identical-component control. Share key, value, and query projections; keep output width fixed; and match recurrent-state size, measured FLOPs, and learned parameter count in separate controlled comparisons. Report the $O(d_{\text{model}}C)$ router parameters and any component-specific transition parameters explicitly, alongside component entropy, token and probability load, effective component count, router gradients, throughput, peak memory, and the usual language-model and retrieval metrics.

13 Conclusion

The exact Gaussian-sum filter and the scalable routed bank belong to one hierarchy. A persistent categorical label gives an exact finite mixture of Kalman associative-memory hypotheses. Allowing the label to change by token yields a conditional mixture of dynamic Bayesian linear regressors, whose exact posterior branches over routing histories. An online structured mean-field projection compresses those histories into C Gaussian bank factors. Restricting the categorical factor to a token-local recognition network gives an amortized filter when the variational objective is trained, and a variationally derived gate when it is not. The resulting fractional Kalman update allocates each token’s information across memories while retaining associative covariance and mean scans. At $C = 1$, the construction reduces to KAM, KDN, or KDA according to the chosen covariance and gain approximation.

Dense soft routing is the reference recurrence but not the capacity-scaling mechanism. A deterministic top- k participation controller, shared key–value coordinates, stable segmented dispatch, and explicitly local-event-time dynamics let recurrent work grow with k while stored state grows with C . Global-time aging is available when skipped transitions can be composed compactly. This separation supports a precise claim: Gaussian-sum KAM supplies the Bayesian motivation, mean-field projection supplies the fixed-size routed update, and sparse segmented scans supply the scalable implementation.

References

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